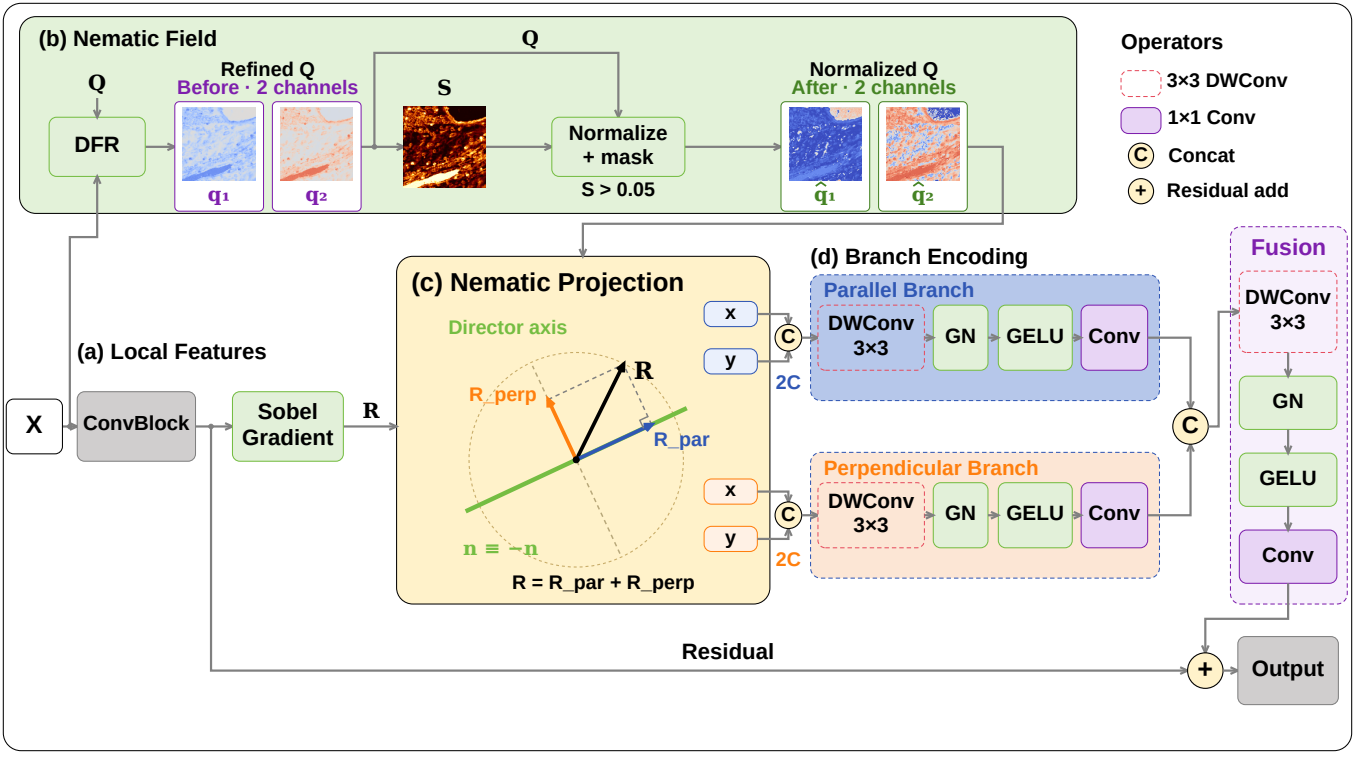

(b) Nematic Field
Q
DFR
Refined Q
Before · 2 channels
q1
q2
S
Q
Normalize
+ mask
S > 0.05
Normalized Q
After · 2 channels
q̂1
q̂2
Operators
3×3 DWConv
1×1 Conv
Concat
Residual add
(a) Local Features
X
ConvBlock
Sobel
Gradient
R
(c) Nematic Projection
Director axis
R
R_perp
R_par
n ≡ −n
R = R_par + R_perp
x
y
2C
(d) Branch Encoding
Parallel Branch
DWConv
3×3
GN
GELU
Conv
Perpendicular Branch
DWConv
3×3
GN
GELU
Conv
C
Fusion
DWConv
3×3
GN
GELU
Conv
Residual
+
Output